

SurfaceLayerDiffusivity: GABLS1 results

Corrected transport comparison | 21 September 2026 | four matched runs, one seed.

Question: can transport confined to one or two near-wall faces improve coarse LES? Four paired GABLS1 runs use WENO9 at 12.5 m resolution (32^3 cells), a 400 m cube, the same frozen source, and 9 h of integration.

Finding: all three SurfaceLayerDiffusivity variants move the mean u , mean θ and w^2 profiles farther from the fixed 1 m LES median by RMS error over native levels below 200 m. This result applies to this grid and these settings; it is not a general verdict on the closure.

Near-wall response: at 12.5 m, final-hour w^2 falls by 85–93%. Skewness changes from +0.56 in the control to -0.21, -0.25 and -0.42. Domain-integrated resolved TKE falls by 15–20%. The first-face variance is suppressed much more strongly than the integrated energy.

Surface exchange: final-hour u^* rises from 0.279 m/s to 0.292, 0.289 and 0.301 m/s, versus a reference mean of 0.262 m/s. Surface cooling also strengthens, moving farther from the reference. One-face 100 and 300 s responses are similar; two-face support increases local suppression.

Smagorinsky comparison: the earlier 12.5 m WENO9 run ($C_s=0.16$, $Pr=1$) has a final-hour peak w^2 of only $1.12e-6 \text{ m}^2/\text{s}^2$, versus 0.0864 in the control. Its mean- u error is smaller despite almost absent resolved turbulence. Teal curves show this historical context; its source differs from the four matched runs. Log axes reveal the contrast.

Next hypothesis: restrict SLD viscosity to u and v , retaining tracer diffusivity and all other settings. This isolates direct damping of w , but indirect effects through shear and stratification remain. No u/v -only result is included. Single-seed comparisons do not establish uncertainty; the LES median is not observational truth.

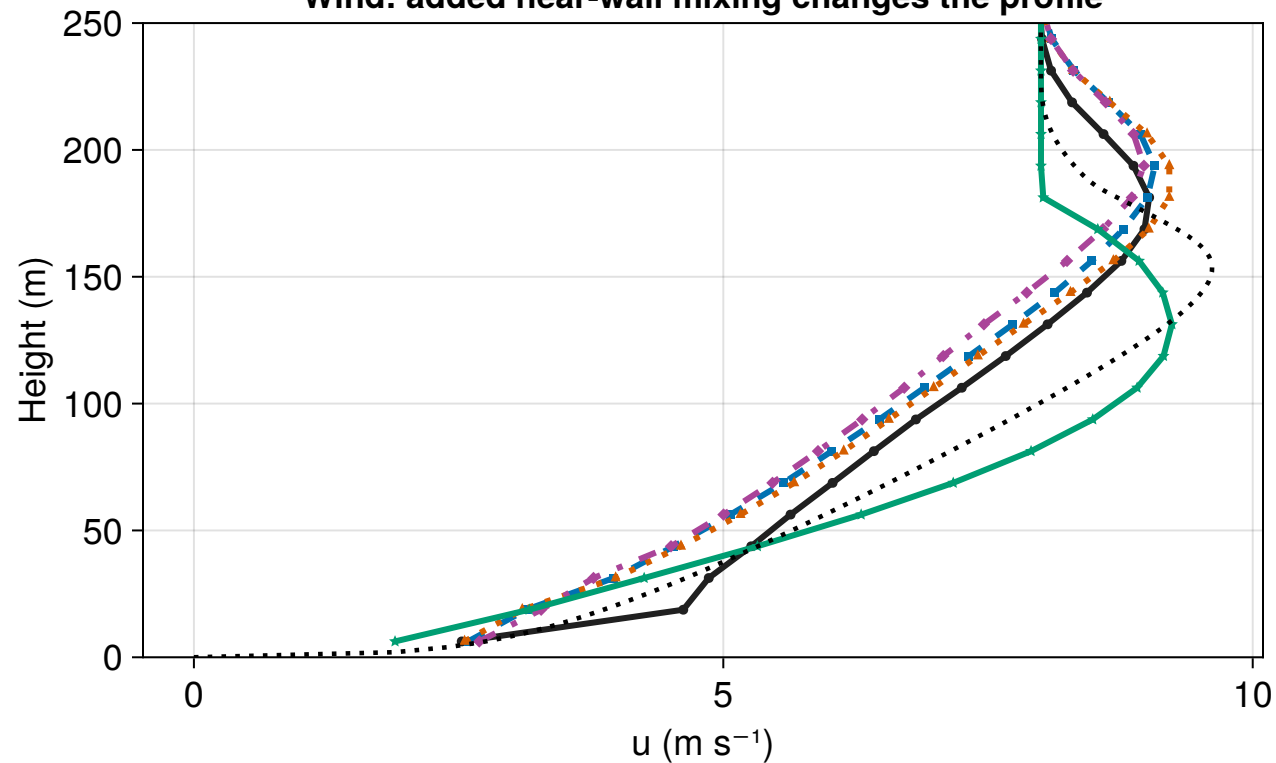
Transport split: corrected diagnostics show SGS mixing supplies 86–95% of first-level u -momentum transport in the SLD cases. Total flux magnitude rises by only 6–17%, while its resolved contribution falls sharply. Mean fields and resolved moments exactly reproduce the earlier runs. The new native coefficient–gradient products pass GPU and export consistency checks; this split does not measure WENO numerical mixing. Historical array 7156 fluxes remain excluded; current figures use corrected array 7293.

Reading the figures: native-height profiles are true 8–9 h means; timelines use instantaneous output. Colors, line styles and markers distinguish cases. Exact data, Julia source, reference hashes and diagnostic exclusions accompany this brief; all original DYCOMS/GABLS1 report material is retained.

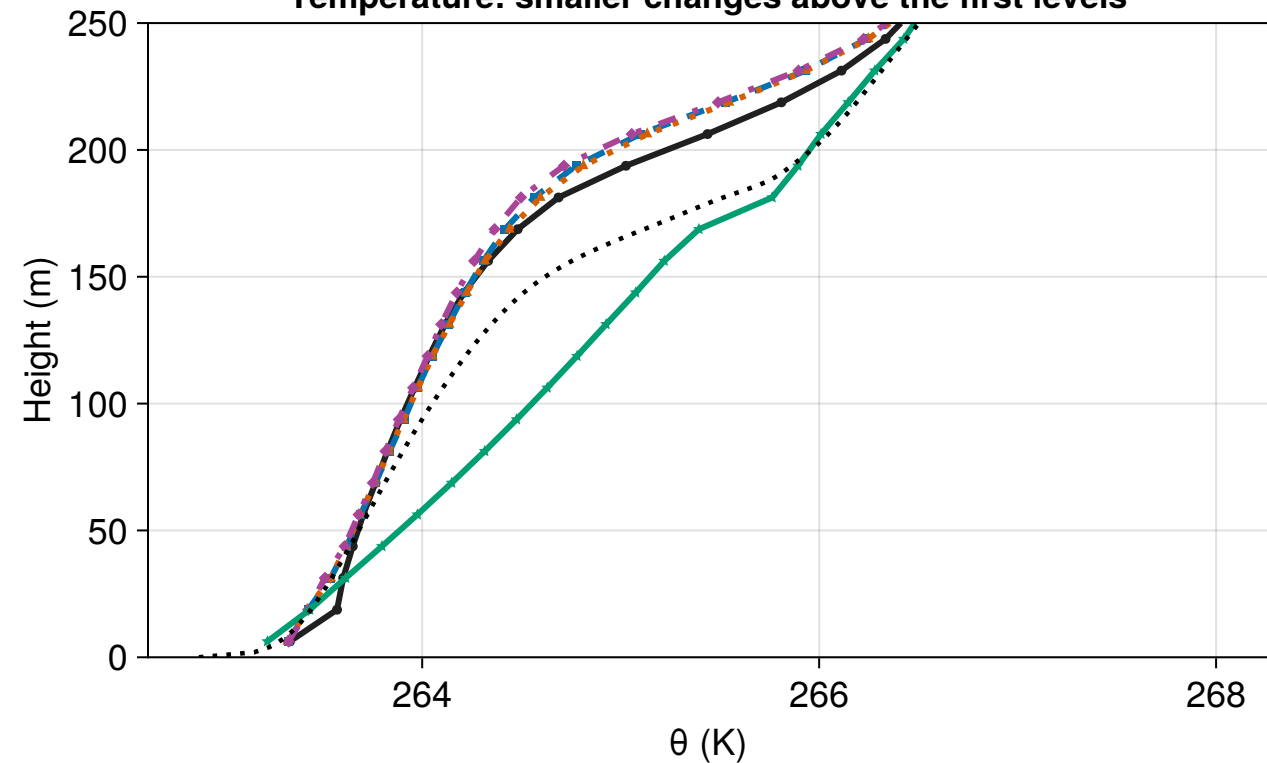
Surface-layer mixing versus coarse-grid Smagorinsky

GABLS1 · WENO9 · 12.5 m grid (32^3) · final hour, 8–9 h · four matched runs + earlier Smagorinsky

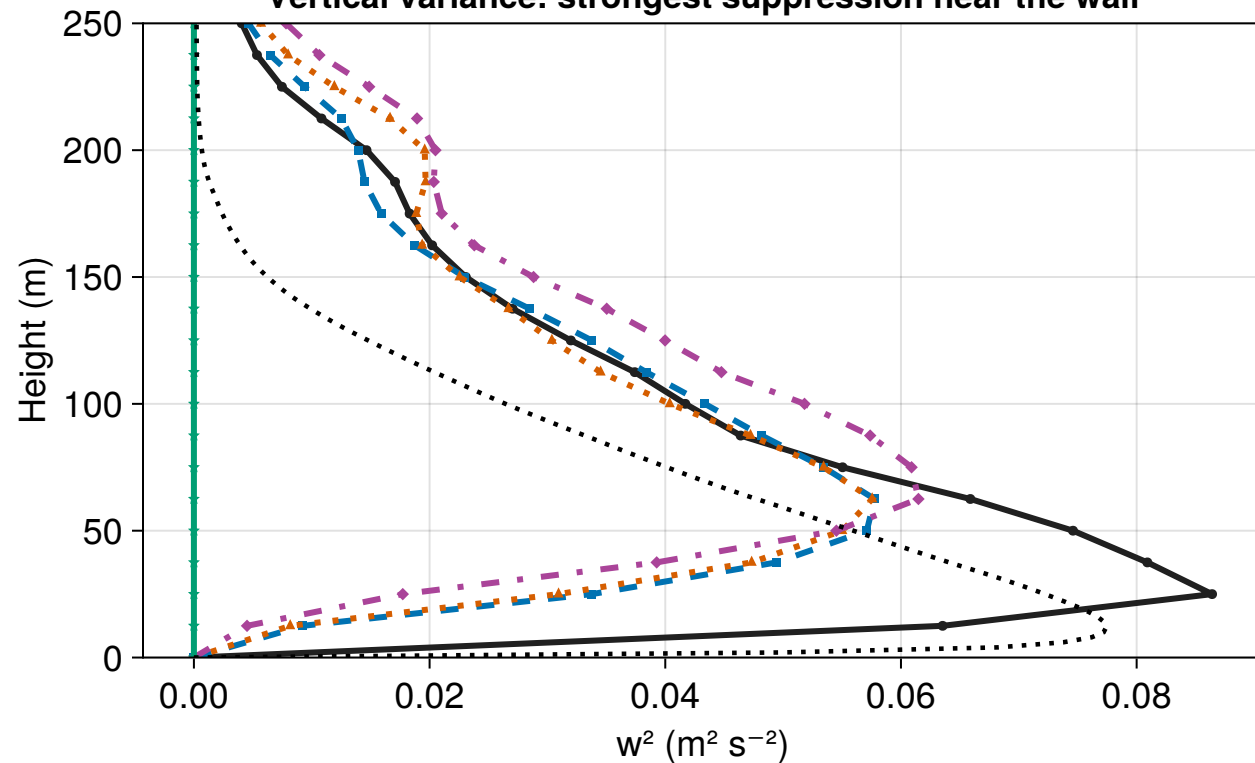
Wind: added near-wall mixing changes the profile



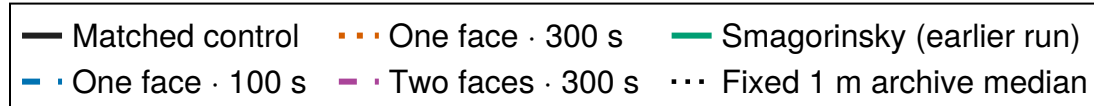
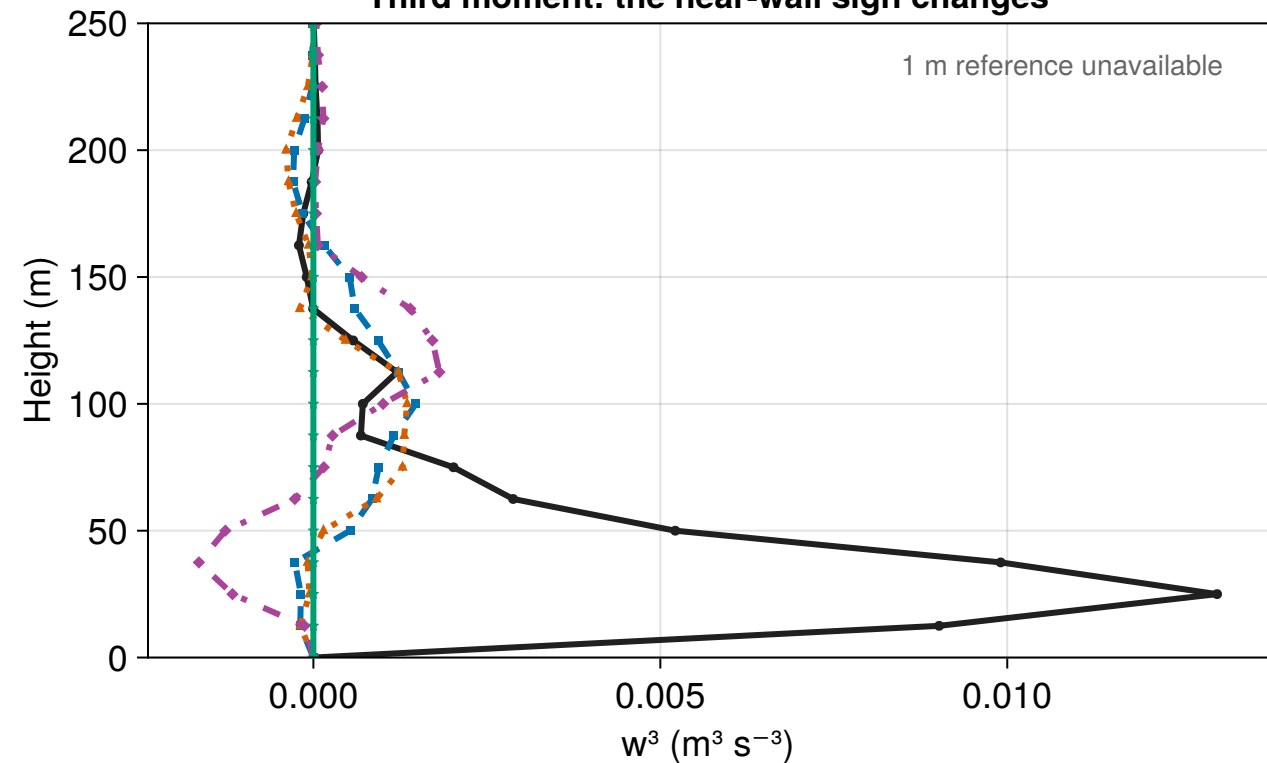
Temperature: smaller changes above the first levels



Vertical variance: strongest suppression near the wall



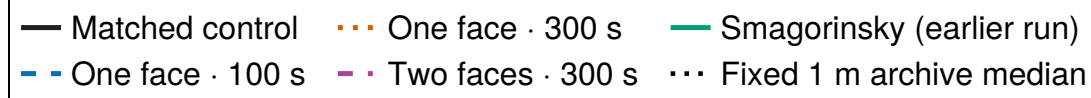
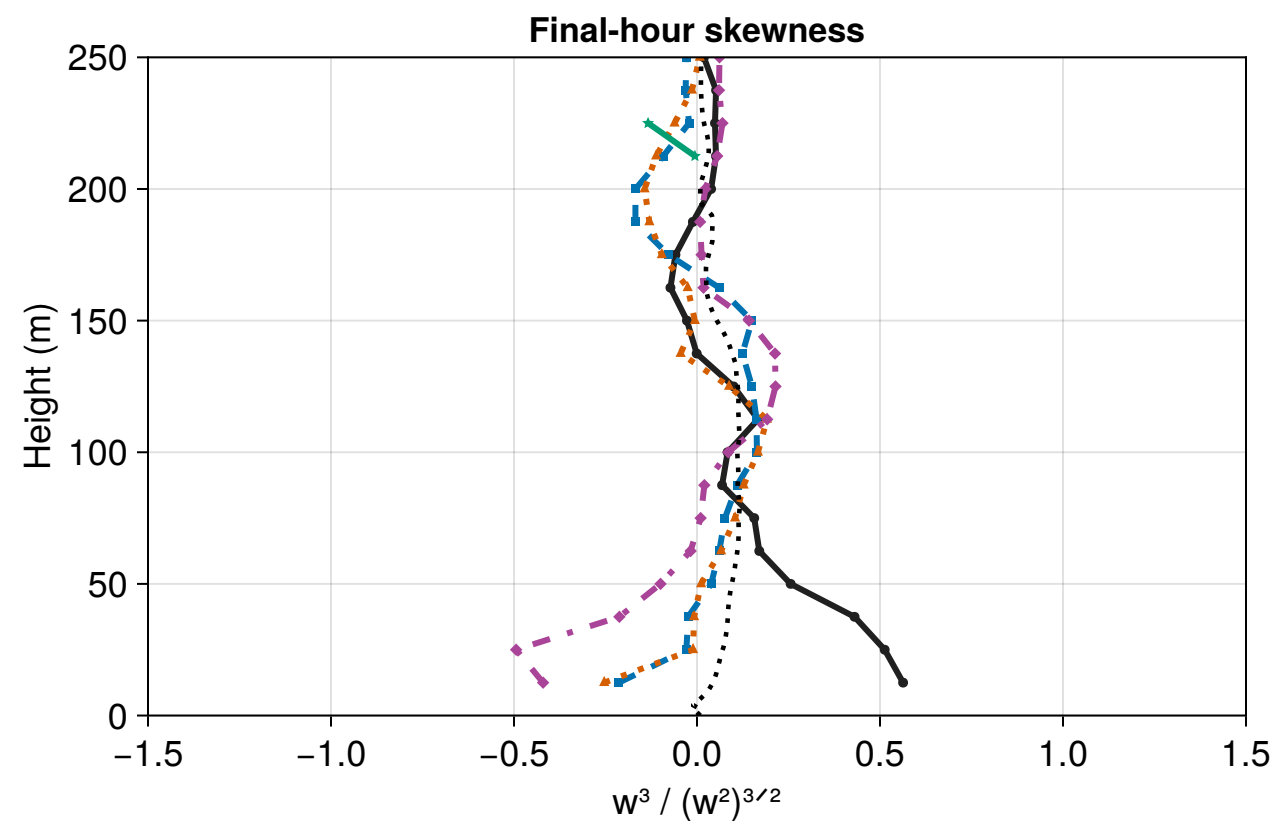
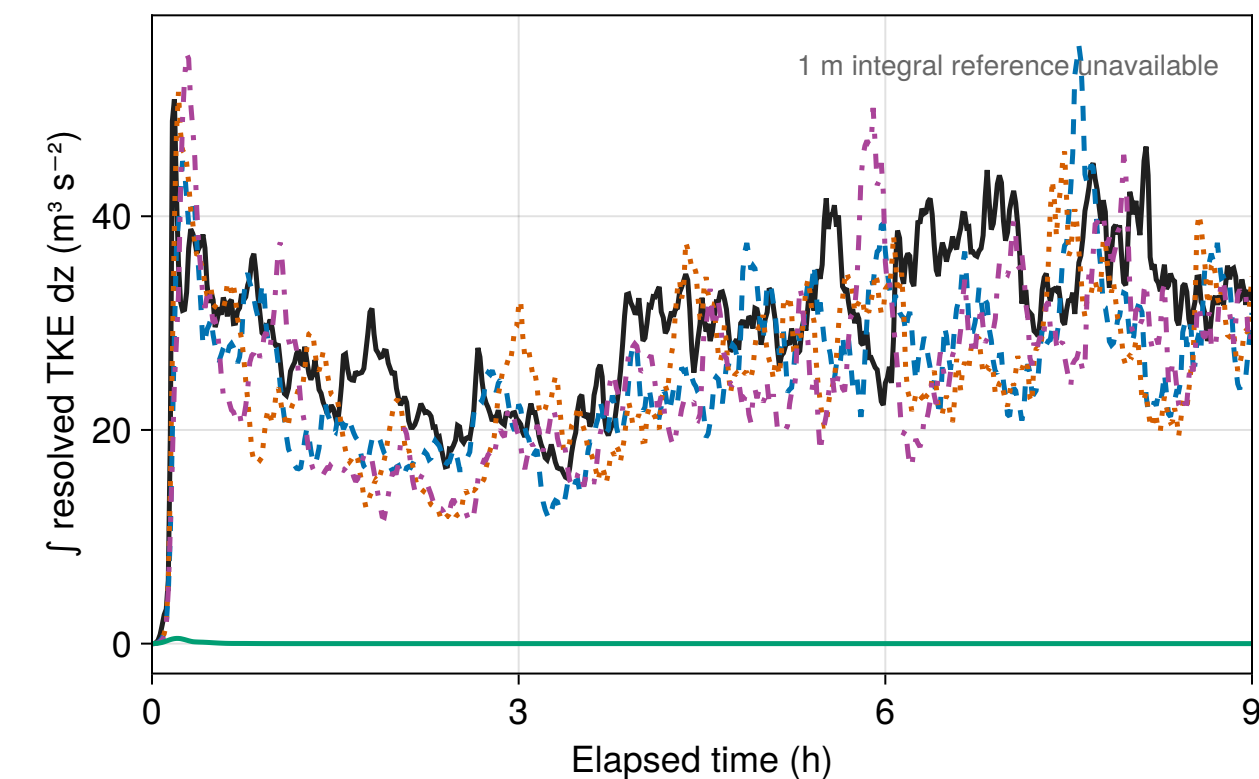
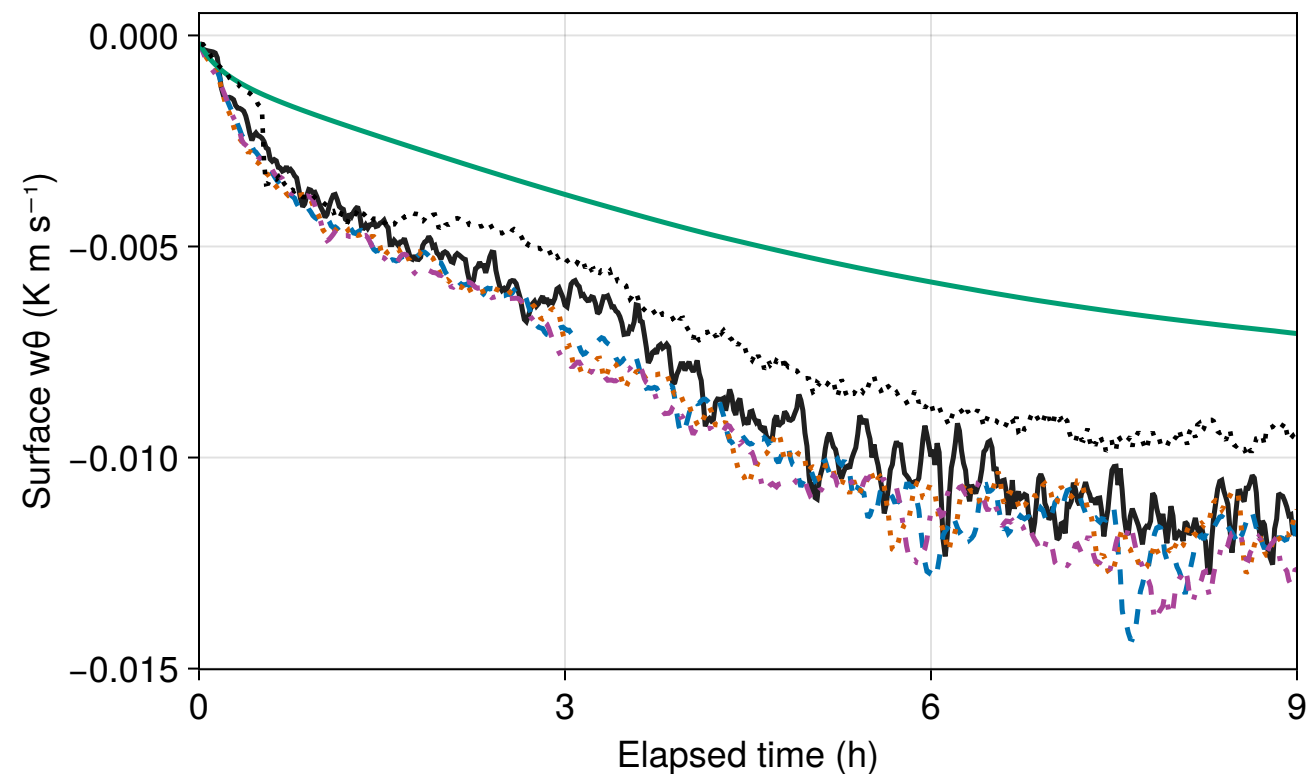
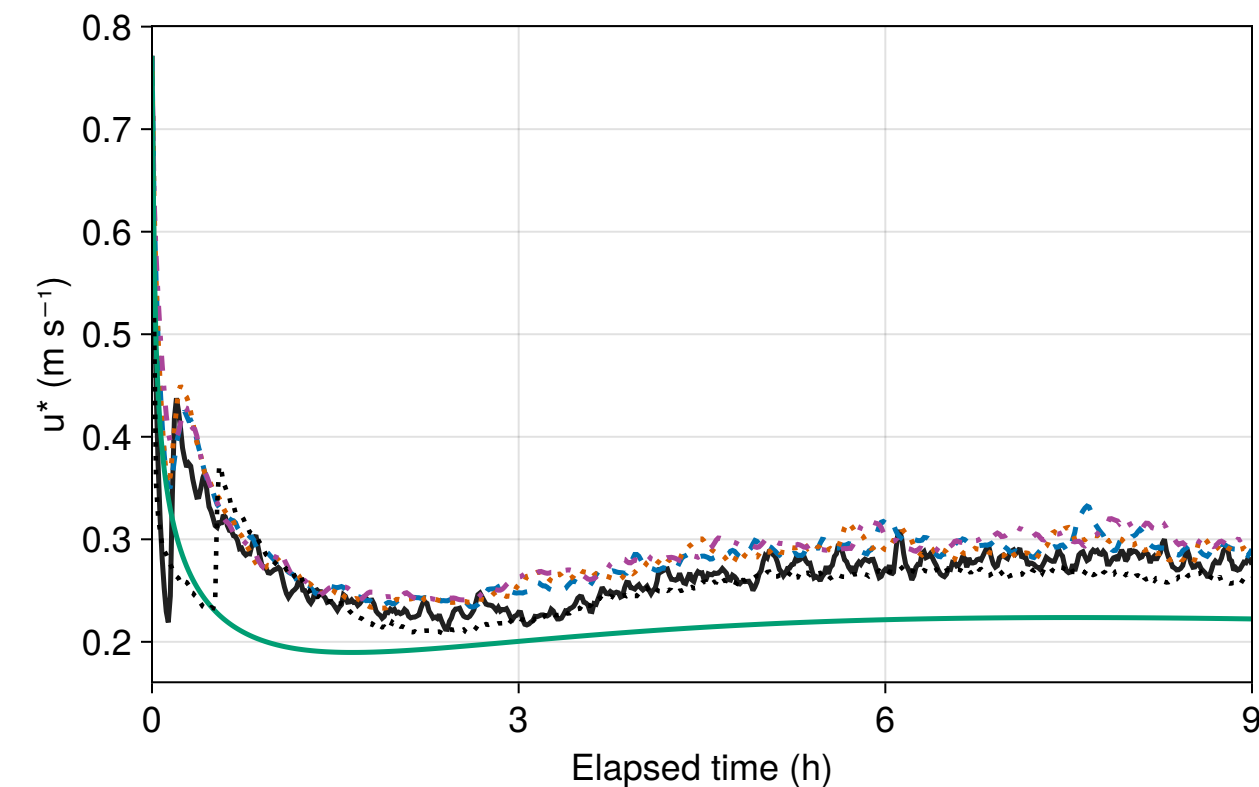
Third moment: the near-wall sign changes



Teal stars: earlier Smagorinsky source; its w^2 and w^3 nearly coincide with zero on these linear axes.
Black dotted: fixed 1 m LES median. Matched runs use the corrected, GPU-validated flux diagnostics.

Follow the response beyond the mean profile

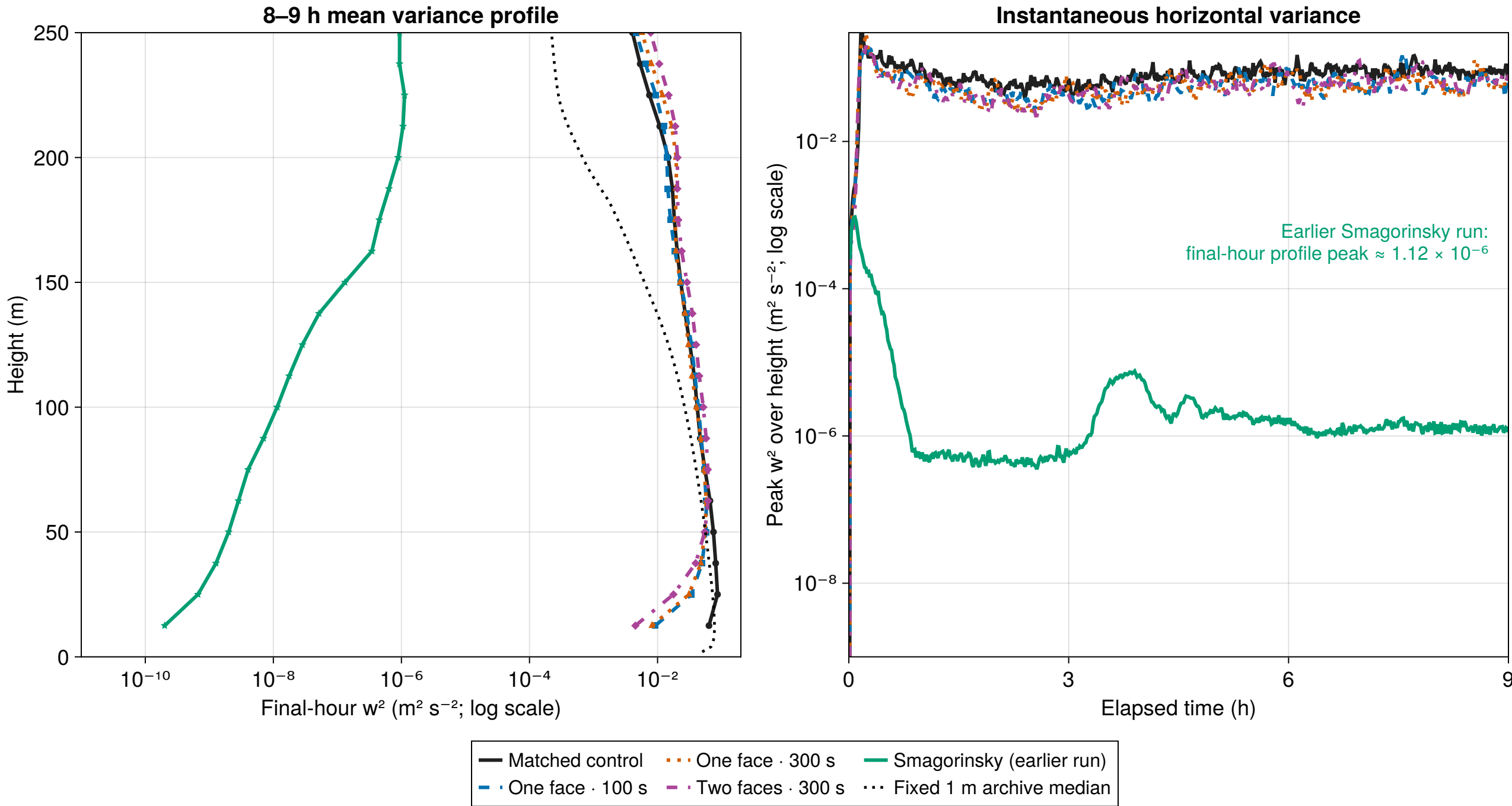
Surface exchange through 9 h; skewness from the final-hour native-face moments



Wall flux remains usable; resolved TKE excludes SGS energy. Teal: earlier Smagorinsky source.
Skewness omits $w^2 < 10^{-6} \text{ m}^2 \text{s}^{-2}$, masking most Smagorinsky levels; a ratio of tiny moments is not robust.

Smagorinsky nearly eliminates resolved vertical turbulence

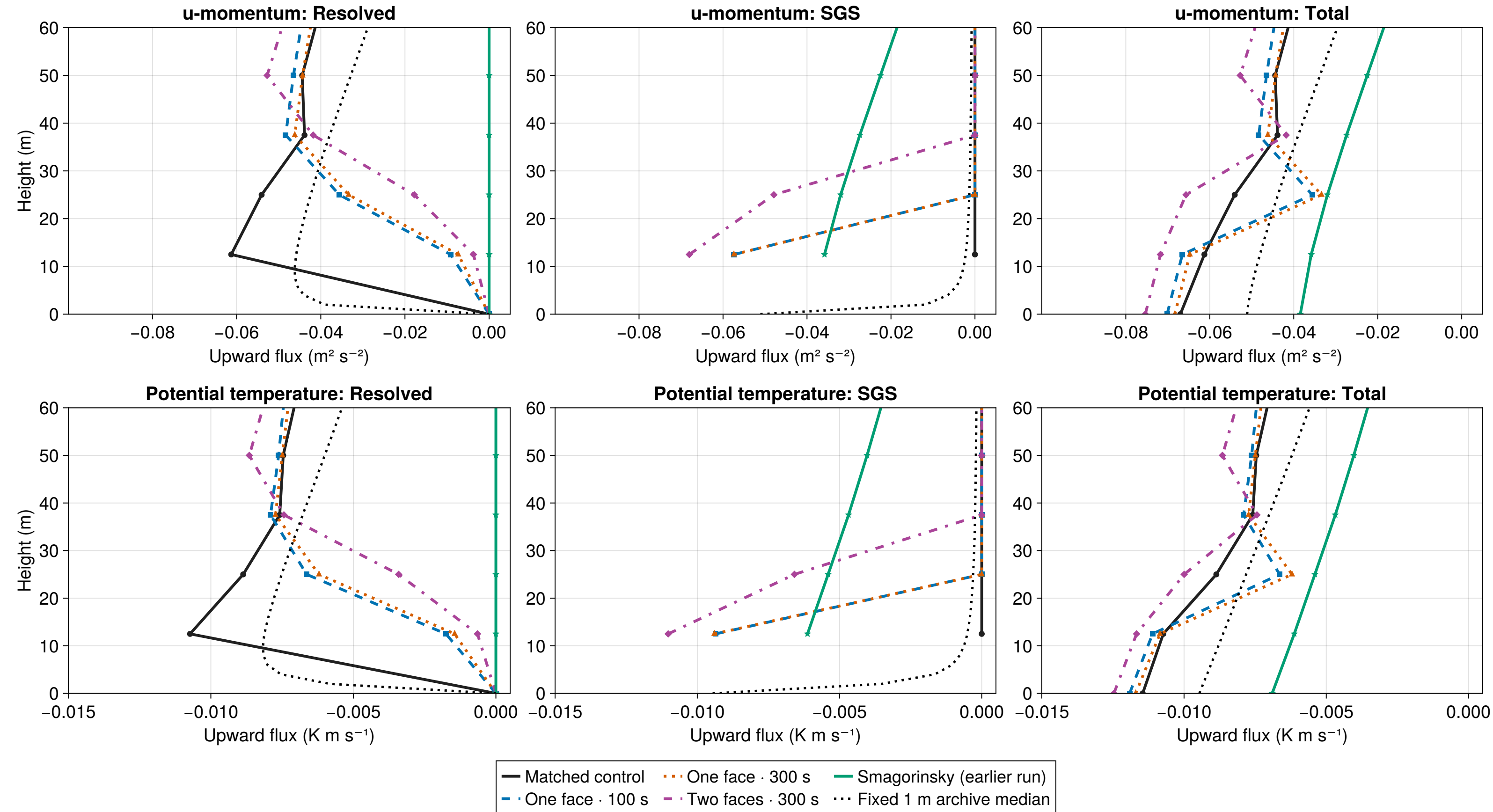
12.5 m GABLS1 · WENO9 · log axes expose the difference hidden by linear scales



Same grid and advection; Smagorinsky is historical context, not a same-revision controlled treatment.
Zeros are omitted on log axes. The left panel averages first; the right panel takes each instantaneous maximum.

SGS transport replaces most of the resolved near-wall flux

GABLS1 · 12.5 m · 8–9 h means · native faces in the lowest 60 m · corrected diagnostics



Colors, line styles and markers identify cases. SGS wall zeros are omitted; total uses the prescribed wall flux. Interior total = resolved + SGS. This split does not quantify WENO numerical mixing. Teal is historical context.

Resolved-flux factor 2: limited response

Matched sensitivity | 21 September 2026 | GABLS1, 12.5 m, WENO9, one interior face, 300 s filter, 9 h, one seed.

Result: crediting twice the resolved transport reduces the added diffusivity, but does not restore resolved turbulence. In the final hour, first-level w^2 rises 7.1%, while peak w^2 falls 6.3% and vertically integrated resolved TKE falls 6.4%. Both factors remain far below the earlier no-closure near-wall variance.

The intervention: factor a multiplies only the signed local filtered resolved flux inside the closure deficit $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$. The same factor is used for momentum and heat. Factor 2 assumes an additional numerical flux equal and aligned with resolved transport. It neither doubles viscosity nor rescales diagnostic fluxes. The experiment does not measure numerical transport or establish that assumption.

Mechanism: final-hour first-face viscosity falls from 1.321 to 1.118 m^2/s (15.4%); heat diffusivity falls from 1.216 to 0.774 m^2/s (36.3%). The first-face SGS share of u-momentum transport remains large: 88.7% versus 86.8%. Reduced closure coefficients therefore produce only a modest change in the resolved/SGS partition.

Moments: at 12.5 m, w^2 changes from 0.008171 to 0.008754 m^2/s^2 , versus 0.06353 without interior closure. The third moment changes from $-1.872\text{e-}4$ to $-1.186\text{e-}4$ m^3/s^3 , and skewness from -0.253 to -0.145. Skewness is the ratio of hourly averaged moments, not an average of instantaneous skewness.

Mean profiles and exchange: u^* falls from 0.2886 to 0.2831 m/s and sensible heat flux changes from -15.54 to -15.38 W/m^2 . Against the fixed 1 m median at native heights $0 < z \leq 200$ m, final-hour RMS errors change by +0.3% for u , -9.2% for θ , and -7.3% for w^2 . Smaller profile errors coexist with lower resolved energy; they do not demonstrate restored turbulence.

Hour sensitivity matters: in 7–8 h, first-level w^2 instead falls 5.1%, integrated TKE falls 10.4%, and the temperature-profile error increases 8.0%. Peak variance and integrated energy decrease in both windows. The near-wall variance increase and temperature improvement in the last hour are not consistent across the two hours. One seed and neighboring hours do not provide an uncertainty estimate.

Comparison integrity: two fresh cases passed strict admission with matching source, grid, seed, initialization and exact 19-profile/541-series schedules. The factor-1 run reproduces every exported profile and time-series value from corrected array 7293's one-face300 case bit for bit. Native physical flux partition, support and coefficients were checked. Fixed 1 m medians are LES references, not observations.

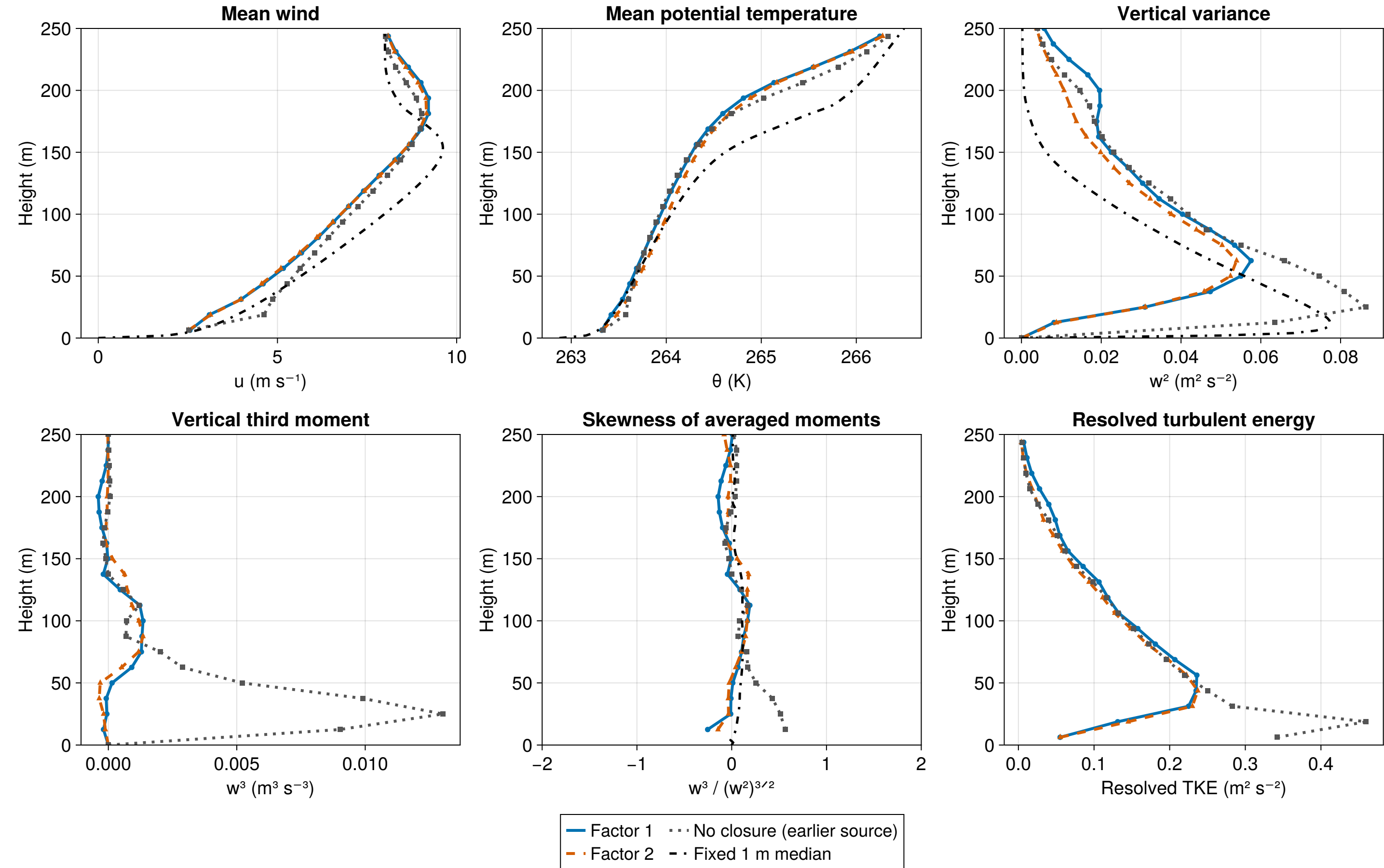
Resolved-flux factor 2: limited response

Execution provenance: both model children completed 32400 s and exited zero. Slurm nevertheless marked the factor-2 batch exit as one. A separate harmless login-shell probe reproduces that pattern through failing `clear_console` in `.bash_logout` under `set-e`; actual batch shell depth was not recorded. The discrepancy and canceled downstream job are preserved. Admission relies on independently complete outputs, durable child exits, matching sources and logs, not a relabeled scheduler success.

Conclusion: this factor-2 test changes coefficients and exchange modestly, but provides no evidence of broad turbulence recovery. It does not identify the numerical flux or establish a calibrated correction factor. All earlier DYCOMS, GABLS1 and SurfaceLayerDiffusivity results remain in the master report. GABLS3's separate humidity-boundary validation is still pending; no new GABLS3 science is included.

Crediting twice the resolved flux: does turbulence recover?

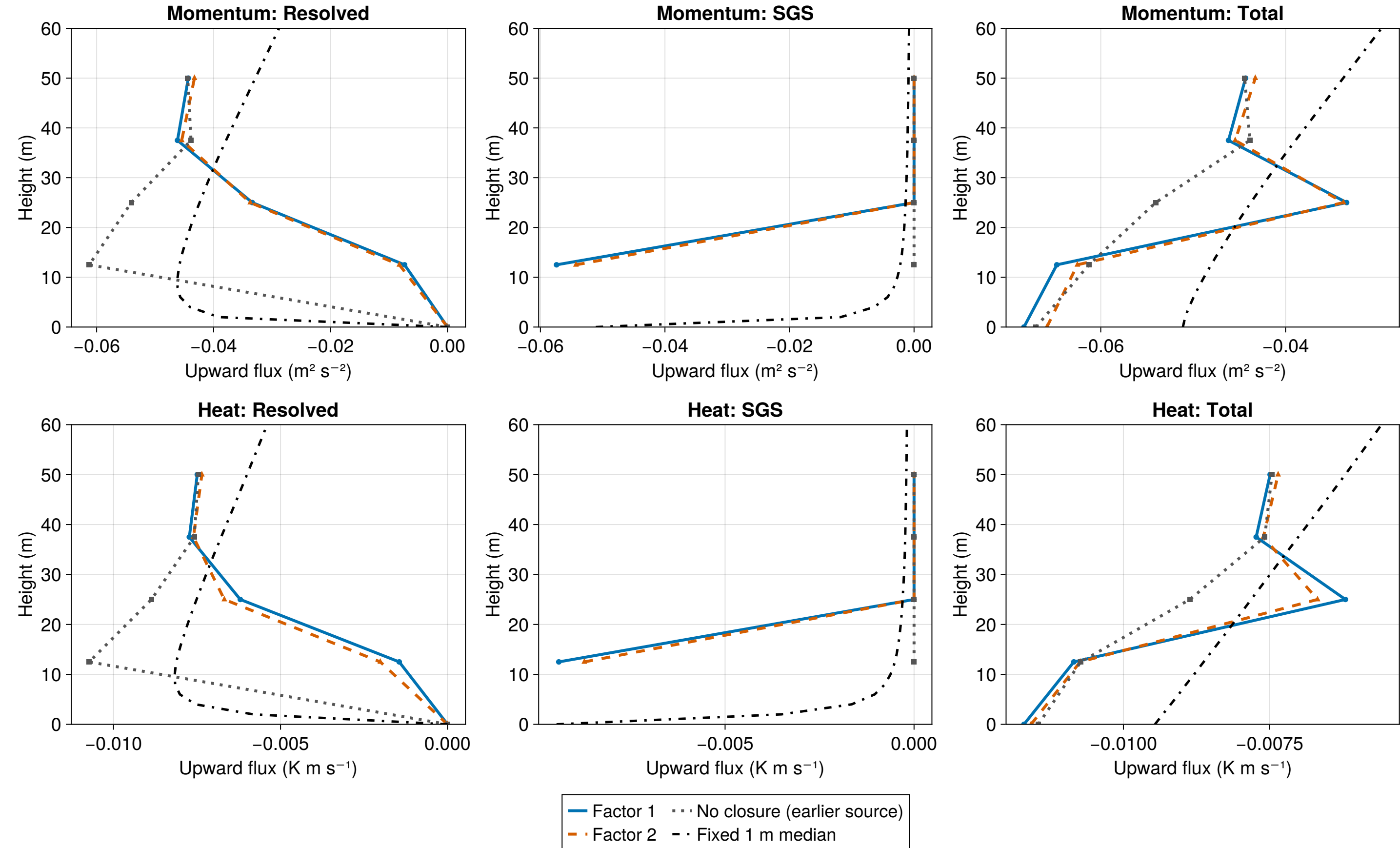
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor changes the closure deficit for momentum and heat; displayed physical fluxes are unscaled.
Earlier no-closure run is context. This tests an assumed numerical contribution, not a measured numerical flux.

Resolved and SGS transport near the surface

Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = resolved + SGS. At the wall, total includes the prescribed flux; artificial SGS wall zeros are omitted.
Each panel includes the fixed 1 m median wherever that reference quantity is available.

Factor 10 mostly switches the closure off

GABLS1 factor-10 sensitivity | 21 September 2026 | 12.5 m, WENO9, one interior face, 300 s filter, 9 h, seed 123. All earlier factor-1/2 results are retained separately.

Result: factor 10 largely removes the added surface-layer mixing after spin-up. Resolved turbulence then resembles the earlier no-closure run. In 8-9 h, first-face w^2 is $0.064671 \text{ m}^2/\text{s}^2$, versus 0.00817056 for factor 1, 0.00875396 for factor 2, and 0.0635343 without closure. Factor 10 is 7.92 times factor 1 and only 1.79% above the no-closure value. This is recovery of resolved turbulence, not proof of accurate total transport.

Mechanism: the multiplier acts only inside the signed local deficit, $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$, for momentum and heat. With $a=10$, aligned resolved transport carrying 10% of the target can satisfy the deficit. Countergradient transport increases it. Crediting ten times the resolved flux assumes a missing contribution nine times as large; the experiment does not measure that contribution. Physical covariance diagnostics are never multiplied.

The saved fractions explain the response. In the final hour, momentum viscosity is exactly zero at 99.9934896% of sampled horizontal point-times; heat diffusivity is zero at 100%. The corresponding raw-zero-deficit and guard-valid-zero-deficit fractions agree. Momentum and heat guards remain valid at 100%, with zero coefficient-cap activity, so this is deficit satisfaction rather than guard failure. Each value averages 60 one-minute samples over all 32×32 horizontal points. It does not imply continuous switch-off between samples: true averaged SGS heat flux is tiny but nonzero, $-1.26136\text{e-}08 \text{ K m/s}$. These six fractions were not saved for factors 1/2 and remain unavailable there.

The closure acts during spin-up. During 0-1 h, mean viscosity is $0.5168 \text{ m}^2/\text{s}$ and heat diffusivity is $0.6870 \text{ m}^2/\text{s}$; zero-coefficient fractions are about 81.54% and 82.98%. By 1-2 h, momentum is zero at 99.9837% of sampled points and heat at 100%. Small intermittent reactivations remain later. In 8-9 h the mean viscosity is $2.72361\text{e-}05 \text{ m}^2/\text{s}$, versus 1.32118 and 1.11797 for factors 1 and 2; sampled mean heat diffusivity is zero.

Moments and energy: factor-10 first-face w^3 is $+0.00977604 \text{ m}^3/\text{s}^3$ and skewness $+0.59443$, compared with negative values for factors 1/2 and $+0.56311$ without closure. Final-hour peak w^2 is 0.0877192 , and integrated resolved TKE is $35.1207 \text{ m}^3/\text{s}^2$, 28.83% above factor 1. The response persists in 7-8 h: first-face w^2 is 0.0668137 , skewness $+0.58543$, and integrated TKE 35.2916 . Neighboring hours and one seed are not an uncertainty estimate.

Factor 10 mostly switches the closure off

Transport and exchange: final-hour SGS carries 0.012852% of first-face u-momentum covariance-plus-SGS transport, versus 88.66% and 86.82% for factors 1/2. Factor-10 u^* is 0.279088 m/s and surface sensible heat flux -15.2328 W/m², close to the no-closure values 0.279164 and -15.2131. The plots' quantity called total is covariance plus actual constitutive SGS. It is NOT the scheme-native WENO advective flux plus SGS: the numerical correction to covariance has not been reconstructed. Agreement of that plotted quantity with a reference cannot validate the missing numerical contribution.

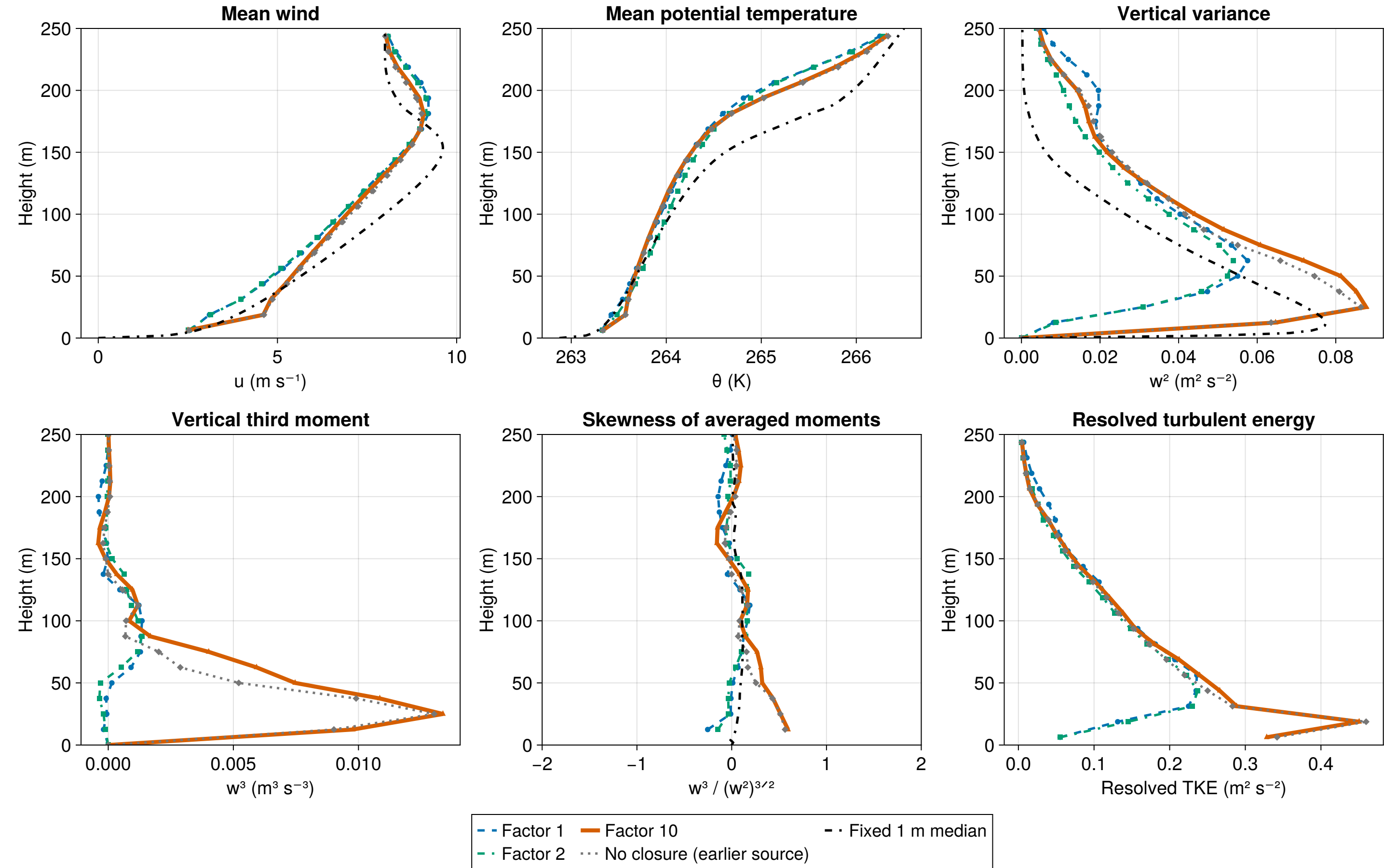
Reference comparison: relative to factor 1, final-hour native-level RMS errors against the fixed 1 m LES median fall 17.33% for u , 9.354% for θ , and 27.43% for w^2 . Nevertheless, factor-10 errors remain larger than the no-closure errors for all three quantities in 8-9 h. The reference is an LES ensemble median, not observations. This strong sensitivity chiefly approaches the no-closure regime; it does not establish factor 10 as a calibrated or faithful correction.

Comparison integrity: factor10 passed separate strict admission (1 admitted, 0 rejected), with 19 native profiles, 541 exact series times, finite values, supported-face SGS and covariance-plus-SGS consistency. Factors 1/2 retain their original admission. Breeze physics commit is identical; factor10's evaluation revision adds read-only diagnostics, with source-specific GPU validation (4,462 checks). Initial native profiles match exactly. Fresh factor1 already reproduces every corrected historical one-face300 profile and series value bit for bit. The earlier no-closure run is labeled as source context. Job7339 used a non-login wrapper and its durable child record verifies exit zero. The preserved older factor2 scheduler discrepancy is documented separately.

Definitions: plotted 8-9 h profiles average the two true half-hour means ending at 30600 and 32400 s; 7-8 h uses 27000 and 28800 s. Series metrics average samples strictly after the hour start through its end. Skewness is the ratio of averaged third moment to averaged variance^(3/2), omitted for $w^2 < 1e-6$. Native model heights are retained; only the fixed 1 m median is interpolated for unweighted RMS errors over $0 < z \leq 200$ m. No WENO flux reconstruction or additional experiment is included.

Factor 10: mean profiles and resolved turbulence

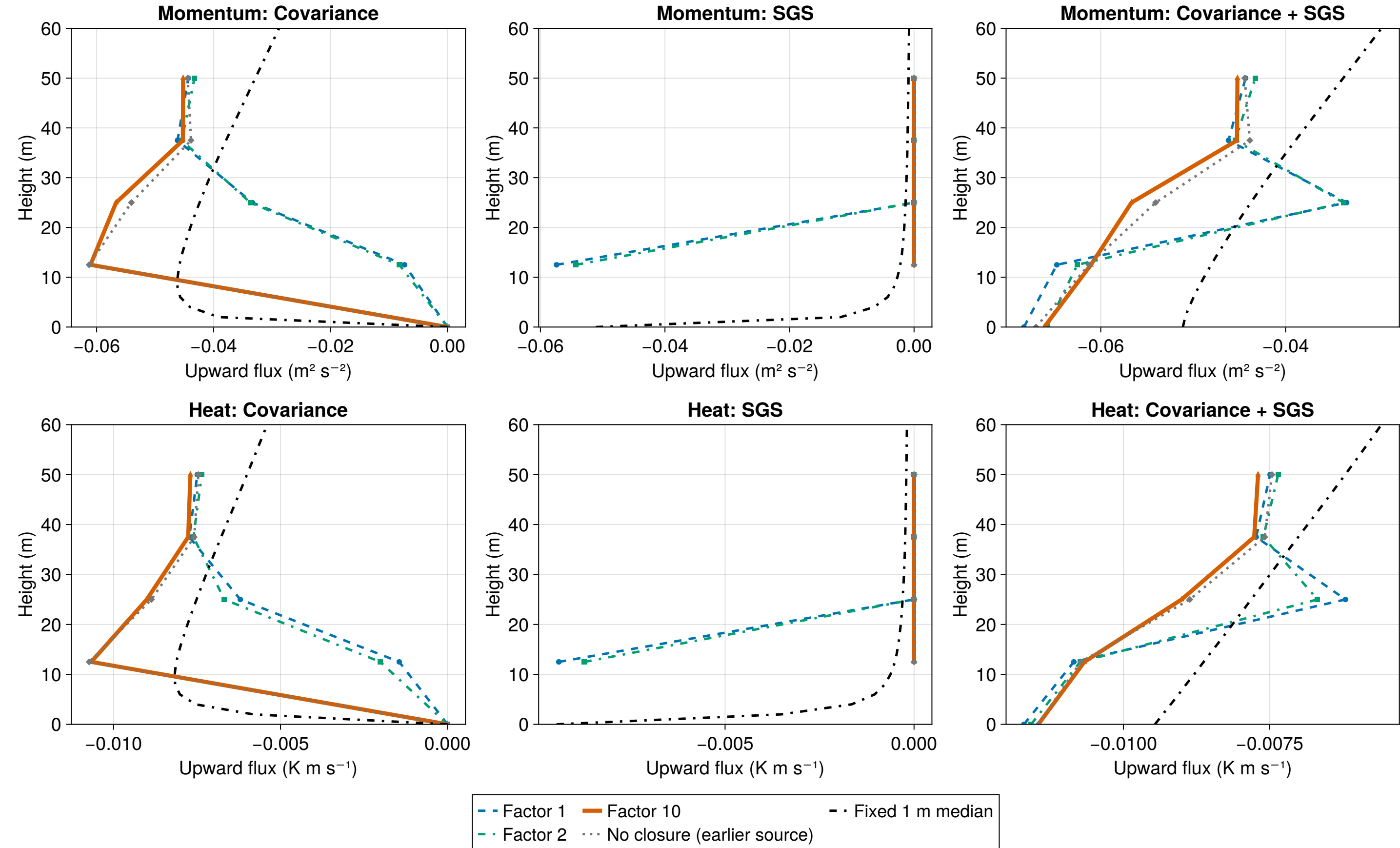
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor 10 is vermillion, solid, triangles. Reference medians shown where available.
Native heights and physical moments retained; the added numerical-flux credit is an assumption.

Covariance and SGS transport near the surface

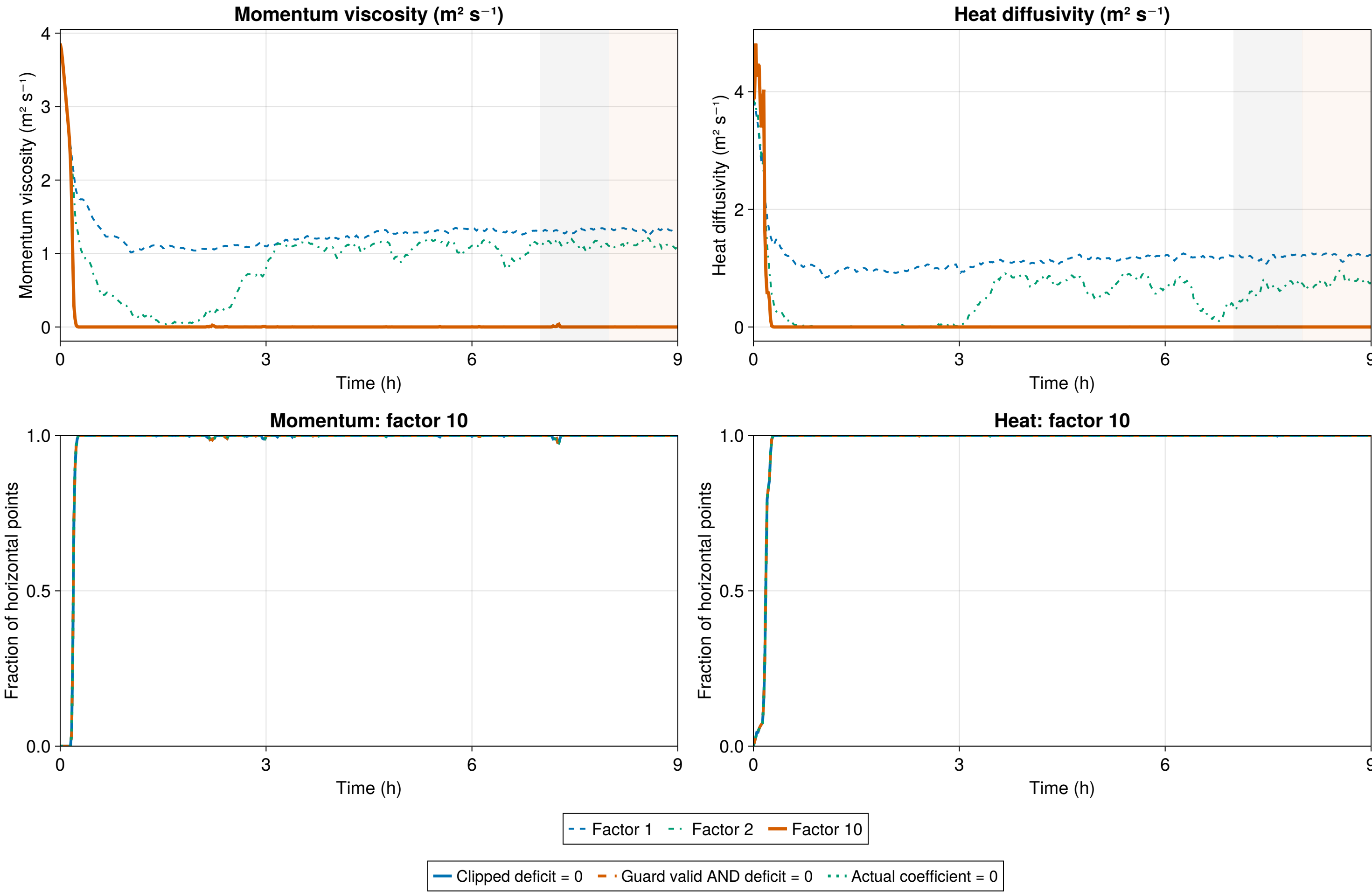
Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = covariance + actual SGS. This is NOT scheme-native WENO transport: its numerical correction is missing.
Wall total is prescribed; artificial SGS wall zeros omitted. Fixed 1 m medians shown where available.

What the factor changes: coefficients and local switch-off

One supported face at 12.5 m · coefficients from all three factors · switch-off fractions saved only for factor 10



All fraction denominators include every horizontal point. Exact overlap is meaningful; quantitative values are in the report.
Active/guard validity is separate from positive mixing. These six fractions were not saved for factors 1 and 2.

Factor 3 mostly switches the closure off

GABLS1 factor-3 sensitivity | 22 September 2026 | 12.5 m, WENO9, one interior face, 300 s filter, 9 h, seed 123. New plots compare only factors 1 and 3, with fixed 1 m LES reference medians where available. Earlier reports and data are preserved.

Result: factor 3 largely switches off the added near-surface closure after spin-up, while resolved turbulence increases strongly. In 8-9 h, first-face w^2 rises from 0.00817056 to 0.060734 m^2/s^2 , a factor of 7.433. Integrated resolved TKE increases 16.38%. First-face w^3 and skewness change from negative to positive. This is a large response to the assumed missing-transport credit, not a calibrated accuracy result.

The intervention remains narrow: multiply signed, locally time-filtered resolved flux by three only inside $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$, for momentum and heat. Aligned resolved transport reaching one-third of the target can eliminate that local deficit; countergradient transport increases it. The assumed extra transport is twice the resolved contribution. Coefficients respond dynamically; neither viscosity nor plotted covariance is simply multiplied by three.

Closure response: final-hour mean first-face viscosity falls from 1.32118 to 0.00417626 m^2/s ; heat diffusivity falls from 1.21593 to 7.01368e-05 m^2/s . Factor-3 viscosity is exactly zero at 98.473307% of sampled horizontal point-times, and heat diffusivity at 99.975586%. The timeline shows substantial initial mixing followed by mostly inactive coefficients, with intermittent reactivation. This is a change in the closure's operating regime, not simply a small uniform reduction in its strength.

Fraction interpretation: each hourly value averages 60 one-minute samples over all 32×32 horizontal points. Raw zero deficit, guard-valid zero deficit and actual zero coefficient are separately retained for momentum and heat; all six fractions appear in the detailed table. All three definitions agree in the final hour. Momentum and heat guards remain valid at 100% and 100%; coefficient-cap fractions are 0 and 0. Thus deficit satisfaction explains the switch-off, rather than invalid targets or active coefficient caps. Sampled zeros do not prove continuous switch-off between outputs. Factor 1 did not save these fractions; its values remain unavailable, not zero.

Moments and hourly sensitivity: factor-3 first-face w^3 is 0.00890504 m^3/s^3 and skewness 0.59496; peak w^2 is 0.0839403 and integrated resolved TKE 31.7259 m^3/s^2 . In 7-8 h, first-face w^2 is 0.0623155, w^3 0.00902423, skewness 0.580117, and integrated TKE 34.7786. The earlier-hour zero-viscosity/zero-heat-diffusivity fractions are 98.18034% / 99.96582%. The two neighboring windows show time dependence; they do not quantify ensemble uncertainty.

Factor 3 mostly switches the closure off

Transport and exchange: the SGS share of first-face u-momentum covariance-plus-SGS transport falls from 88.6586% with factor 1 to 1.14829% with factor 3. Factor-3 u^* is 0.272265 m/s and sensible heat flux -14.5912 W/m², versus 0.288603 m/s and -15.5382 W/m² for factor 1. Every plotted interior total is covariance plus actual constitutive SGS. It is NOT scheme-native WENO advective transport plus SGS: the numerical correction to covariance remains unmeasured. The factor experiment cannot identify or validate that missing flux.

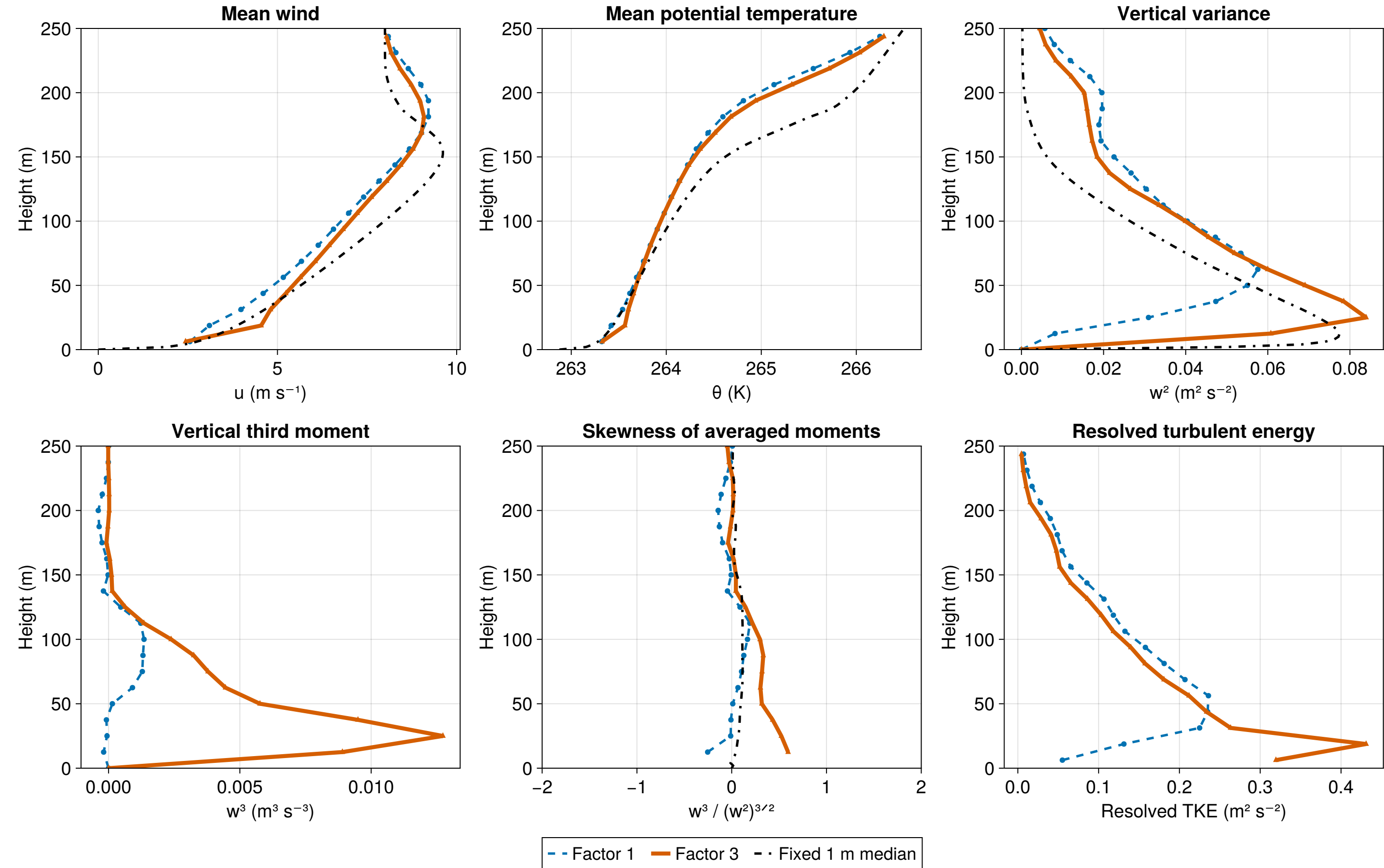
Reference comparison: at native heights $0 < z \leq 200$ m, factor-3 final-hour RMS errors against the fixed 1 m LES median are 0.720275 m/s for u , 0.376886 K for θ , and 0.0132648 m²/s² for w^2 . Changes relative to factor 1 are -23.19%, -9.761%, and -45.94%. The reference is an LES median, not observations. Greater resolved variance or a smaller mean-profile error alone does not establish faithful total transport.

Comparison integrity: factor 3 passed its source-specific strict admission (1 admitted, 0 rejected), with identical Breeze physics and initial native profiles, 19 exact profile times and 541 exact series times. Checksums, finite values, native vertical locations, one-face support, covariance-plus-SGS consistency and fraction bounds were verified. Factors 1/2/10 retain their original admissions. Factor 1 reproduces the corrected historical one-face300 output bit for bit; the no-closure case remains labeled as earlier-source context. The durable child record for non-login job 7348 verifies exit zero; no unavailable batch status is inferred.

Definitions: 8-9 h profiles average true half-hour means ending at 30600 and 32400 s; 7-8 h uses 27000 and 28800 s. Series averages exclude the hour start and include its end. Skewness is averaged w^3 divided by averaged w^2 to the three-halves power, masked for $w^2 < 1e-6$. Native model heights are retained; only the reference is interpolated for unweighted RMS errors. No scheme-native flux reconstruction, new closure design, or additional simulation is included.

Factor 3: mean profiles and resolved turbulence

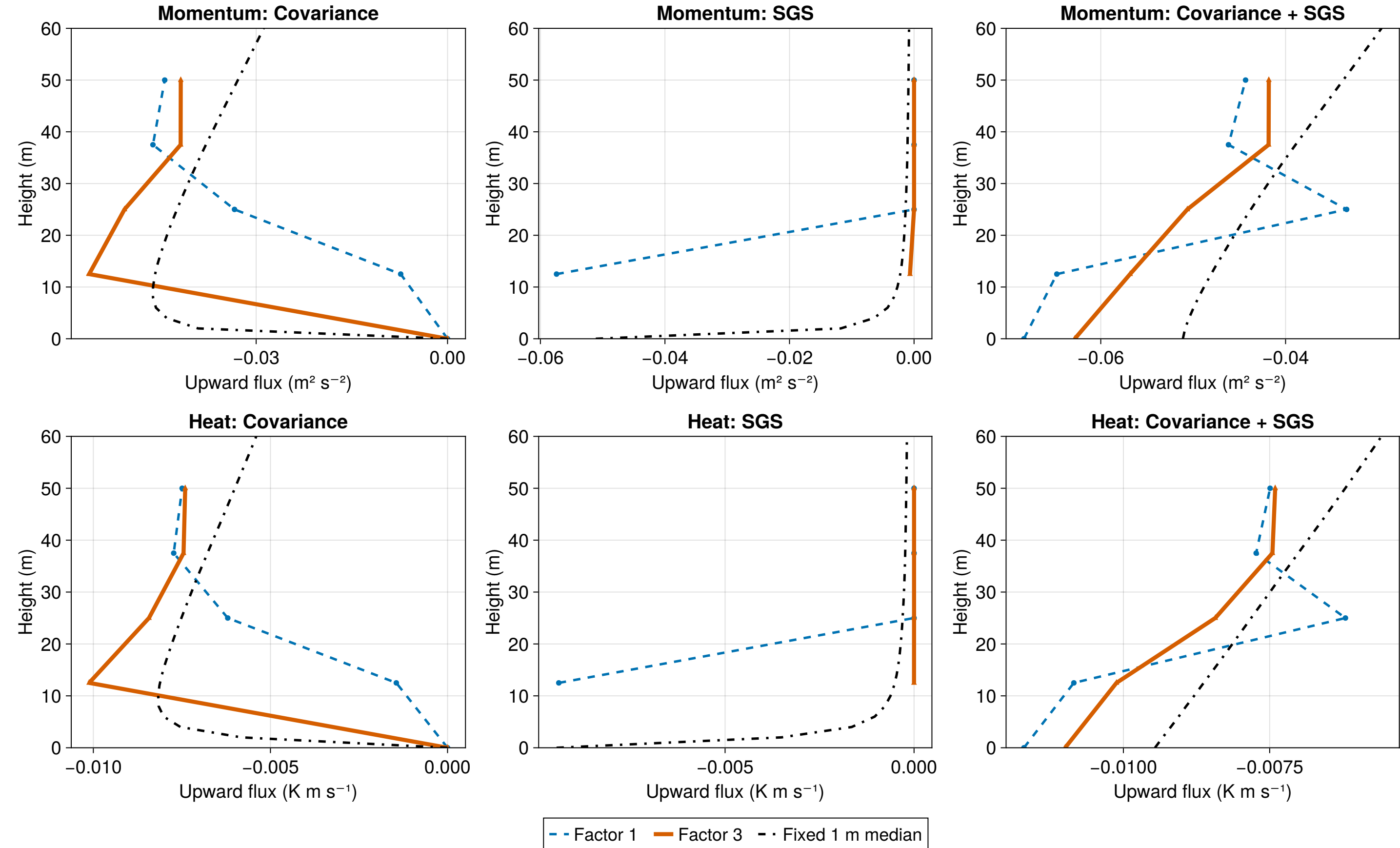
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor 1 is blue, dashed, circles; factor 3 is vermillion, solid, triangles.
Native heights and physical moments retained. Fixed 1 m reference medians shown where available.

Covariance and SGS transport near the surface

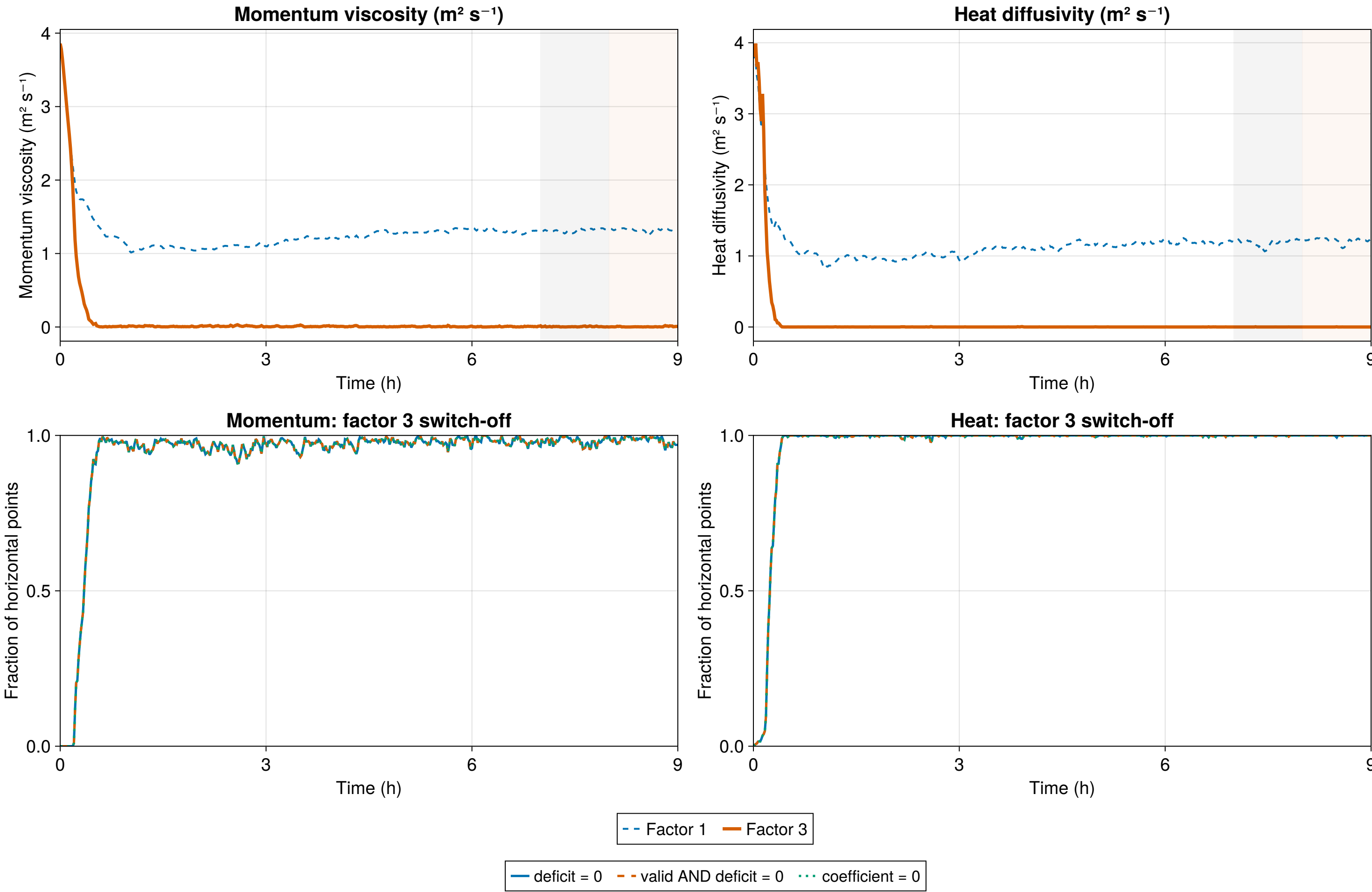
Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = covariance + actual SGS. This is NOT scheme-native WENO transport: its numerical correction is missing.
Wall total is prescribed; artificial SGS wall zeros omitted. Fixed 1 m medians shown where available.

What the factor changes: coefficients and local switch-off

One supported face at 12.5 m · factors 1 and 3 · factor 1 switch-off fractions unavailable



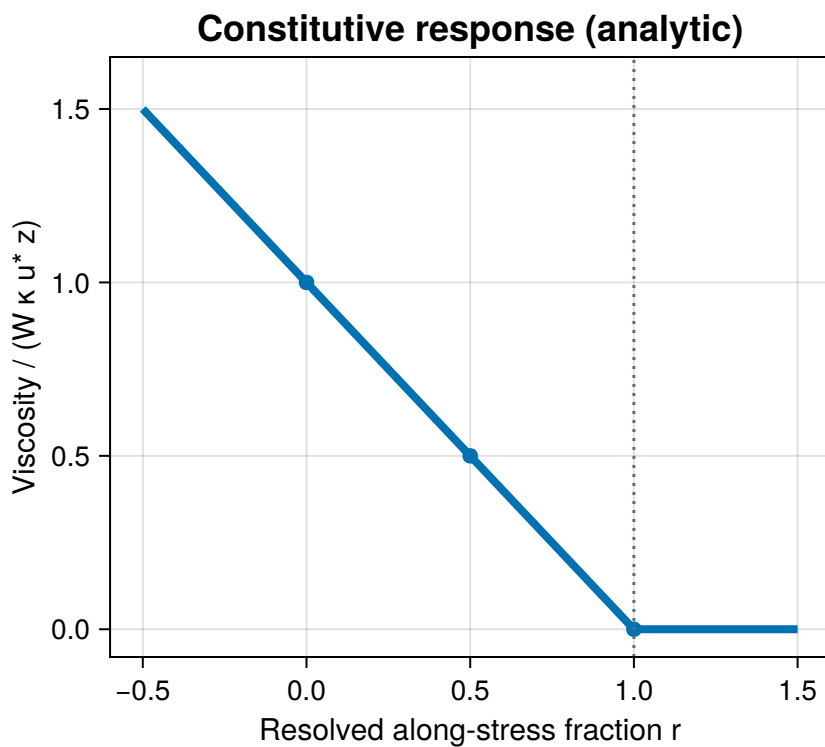
All fraction denominators include every horizontal point. Exact overlap is meaningful; quantitative values are in the report.
Active/guard validity is separate from positive mixing. These six fractions were not saved for factor 1.

Can a thin surface closure improve coarse LES?

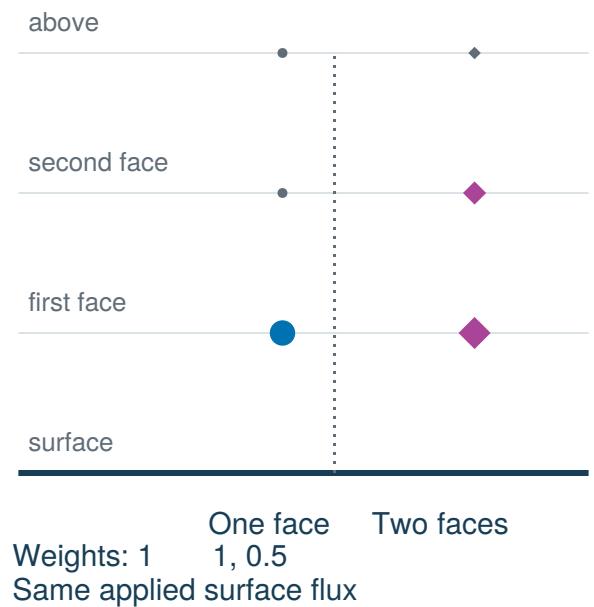
Supply unresolved near-wall transport while preserving resolved turbulence.
Four coarse GABLS1 comparisons admitted; transfer tests ongoing.

A local estimate of missing transport

Filter velocity, scalar, and wall-flux samples in physical time. Use their centered covariance to estimate the resolved share of the surface flux.



Shallow vertical support



What this formula does

No resolved stress: full neutral similarity viscosity. Half resolved: half the viscosity. Fully resolved or more: the correction turns off.
Opposite-sign transport can give a factor greater than one.

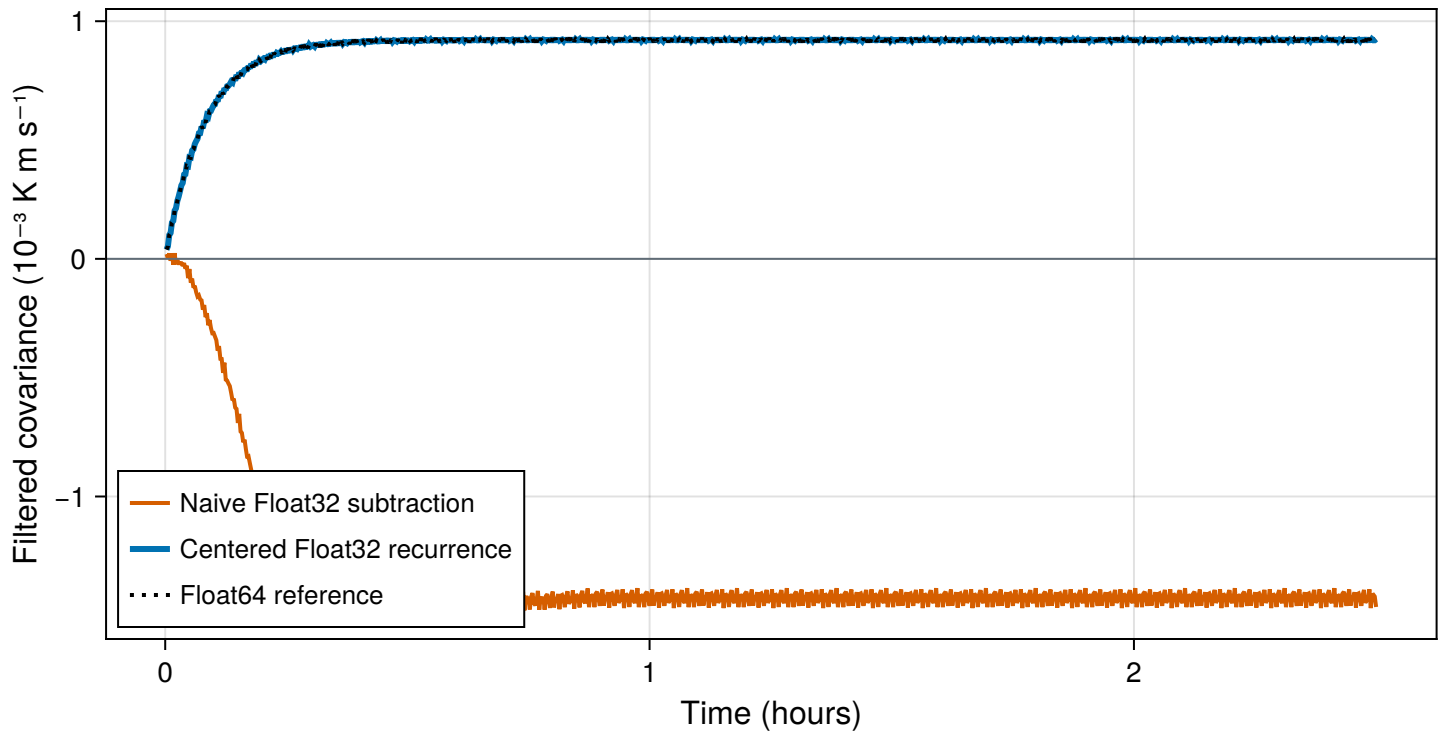
What it does not establish

The added flux still depends on the instantaneous vertical gradient.
The formula does not force exact flux balance, include stability functions, or measure the numerical transport produced by WENO.

One-face height: 12.5 m in both planned GABLS comparisons.
Scalar correction uses signed flux ratios; zero/tiny wall flux disables it.

A small covariance can get the wrong sign in Float32

Synthetic arithmetic test: 90,000 samples, $\Delta t = 0.1$ s, filter time = 300 s.
Identical Float32 input samples are used in the Float64 reference.



Final covariance: naive -1.465 | centered 0.925 | reference 0.924

Units: $10^{-3} \text{ K m s}^{-1}$. Centered recurrence relative error: 0.020%.

Why this matters for the closure

Subtracting two much larger filtered products loses the small turbulent flux. Here it reverses its sign. A centered covariance update retains the signal that controls the added diffusivity.

Implementation checks support the arithmetic result

The original 119-check CPU suite covers filtering, signs, guards and conservation. Later halo/restart checks and the 1,678-check GPU suite support the admitted run source. Their evidence files retain each revision.

Boundary of the evidence: this arithmetic example is synthetic.
Physical fidelity is assessed separately using the completed GABLS1 runs.
Reproducer, exact plotted CSV, and source hashes accompany this section.

Judge transport, turbulence, and the mean state together

One controlled intervention, three complementary tests of physical behavior. See the accompanying GABLS1 results for the first completed comparison.

01 Did the near-wall transport change as intended?

Lead with resolved, added, and combined momentum/heat flux profiles. Zoom the lowest four levels; mark every active face. Pair flux mismatch with viscosity, diffusivity, guard activity, and mean vertical transport.

02 Did the boundary layer improve without losing turbulence?

Place mean wind and temperature beside w^2 , w^3 and skewness. Show surface flux through time; withhold invalid depth diagnostics. Retain the fixed 1 m GABLS1 median; missing reference quantities remain explicit.

03 Is the result robust to filter time and support?

Use the same four styles in every GABLS panel. Compare with a freshly run, same-revision control, then examine 7-8 h versus 8-9 h. Report changes in physical units; avoid a composite score that hides tradeoffs.

-  Matched WENO9 control
-  One face / 100 s
-  One face / 300 s
-  Two faces / 300 s

Independent challenges strengthen the conclusion

GABLS3: moist nocturnal evolution and the sunrise flux sign change.
Neutral ABL: fixed-stress transport, without a surface-drag prediction.
Both add evidence; neither substitutes for the GABLS1 reference comparison.